Diagrammatic Illustration

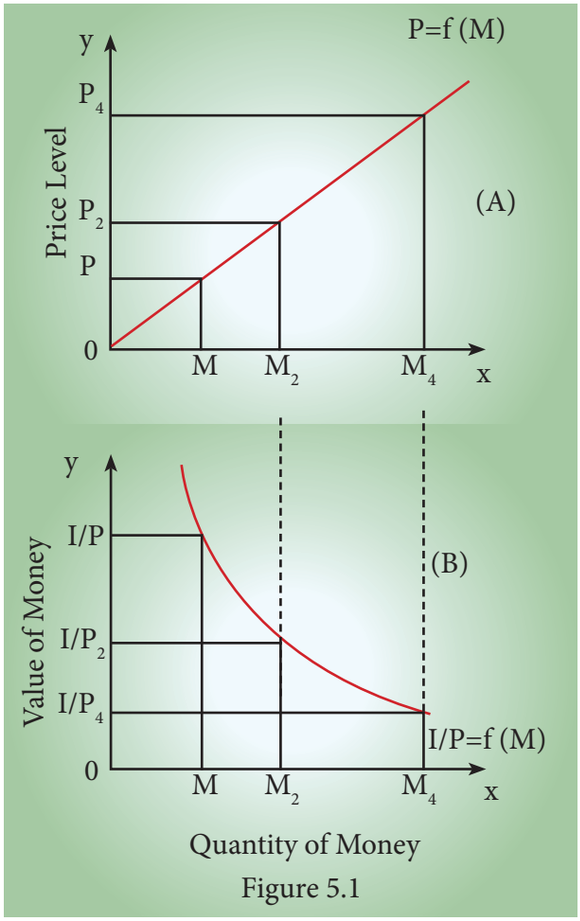


Figure 5.1

Figure (A) shows the effect of changes in the quantity of money on the price level. When the quantity of money is OM, the price level is OP. When the quantity of money is doubled to OM_2, the price level is also doubled to OP_2. Further, when the quantity of money is increased four-fold to OM_4, the price level also increases by four times to OP_4. This relationship is expressed by the curve OP = f (M) from the origin at 45^0.

Figure (B), shows the inverse relation between the quantity of money and the value of money, where the value of money is taken on the vertical axis. When the quantity of money is OM, the value of money is OI / P. But with the doubling of the quantity of money to OM_2, the value of money becomes one-half of what it was before, (OI / P_2). But, with the quantity of money increasing by four-fold to OM_4, the value of money is reduced by OI / P_4. This inverse relationship between the quantity of money and the value of money is shown by downward sloping curve IO / P = f(M).

b) Cambridge Approach (Cash Balances Approach)

i) Marshall's Equation

The Marshall equation is expressed as:

M = KPY

Where

M is the quantity of money

Y is the aggregate real income of the community

P is Purchasing Power of money

K represents the fraction of the real income which the public desires to hold in the form of money.

Thus, the price level P = M/KY or the value of money (The reciprocal of price level) is 1/P = KY/M

The value of money in terms of this equation can be found out by dividing the total quantity of goods which the public desires to holdout of the total income by the total supply of money.

According to Marshall's equation, the value of money is influenced not only by changes in M, but also by changes in K.

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